Intro

It's time to show students what they'll be creating this week!

Click **here** to play Ledge Throw!

How to Play

•Use arrow keys to move left / right
•Use the up key to jump
•Use the space key to throw a ledge in the direction you're facing

Objectives

•Reach the door at the end of the level

Current Code

What code is currently in our game?

Player

Setup

This code sets the gravity (how fast we fall down), makes sure our Player can't leave the world, and has code to allow touch users to play

Right

If we try to move right, this code will get called. It sets the facingRight variable to true, sets the speed of the Player to 200 and makes sure the Player looks right.

Left

The same concept as the Right function. This code will set the facingRight variable to false and play the left animation and move to the left at a speed of 200.

Ledge

Setup

We need to setup a few things for our Ledge actor. This code sets the value of x to 0 and y to it's starting **y** position. Since we don't want our Ledge to fall through the ground, it's not affected by gravity. No rotation ensures it doesn't ever turn / flip itself unexpectedly. Finally, making it a sensor ensures that it can't collide with any tiles or our Player.

Door

Next level

Self explanatory! This function takes us to the next level (when our Player hits it).

Lesson Plan

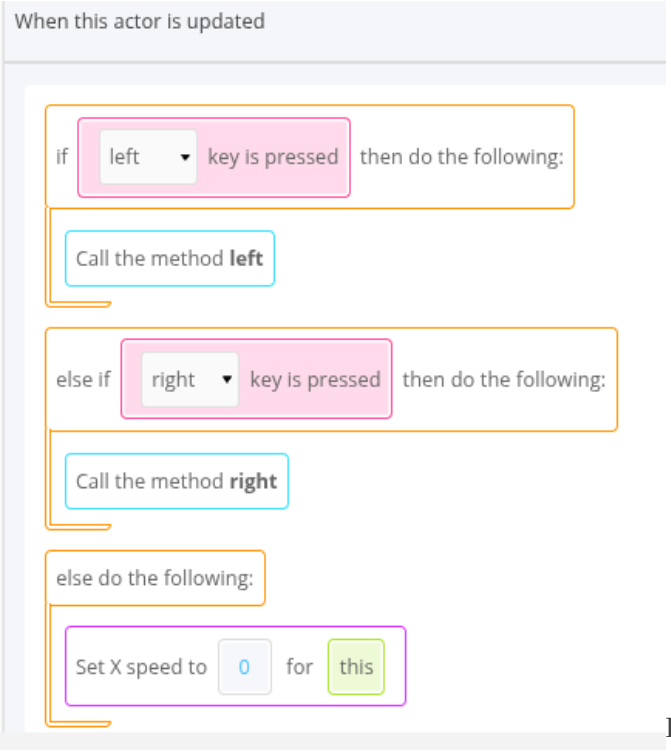
Ledge Throw

Oh no!! We're trapped! We need to rescue ourselves by throwing and jumping on ledges to reach the end of the level. We'll be `counting` actors, creating actors, and learning how to limit jumps with a `jumpCount` variable!

1. Define **movement** for our player

Click on menu: Ledger Throw

Click on Player, click on code tab, add new function



Hit save and test it out! Make sure your left and right movement works!

2. Define the JUMP!!

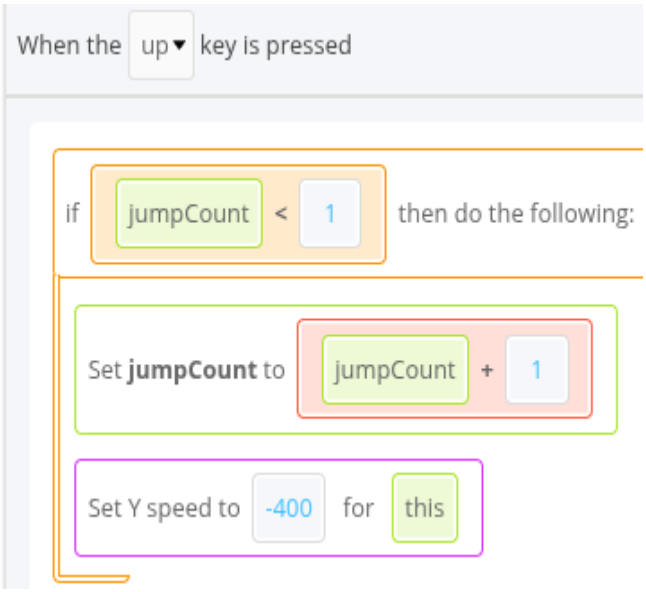
Even though jumping is really just changing the y speed, this function will have a bit more in it! Why? Because we want to be able to **limit** the number of jumps the Player can do. How EASY would it be if the Player could just jump to the top of the level?? That'd be cheating!

Rather than letting the game allow us to jump forever, let's make it keep track of how many jumps we've done.

Click on menu: Ledger Throw

Click on Player, click on code tab, add a new function `jump`.

Add a `jumpCount` variable

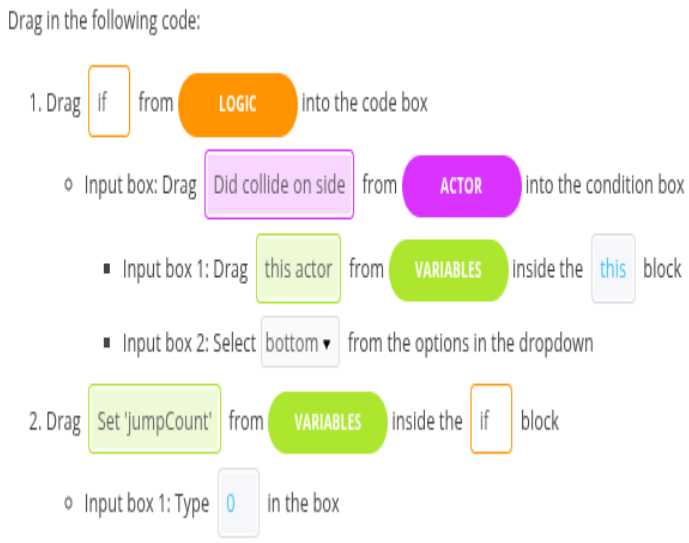


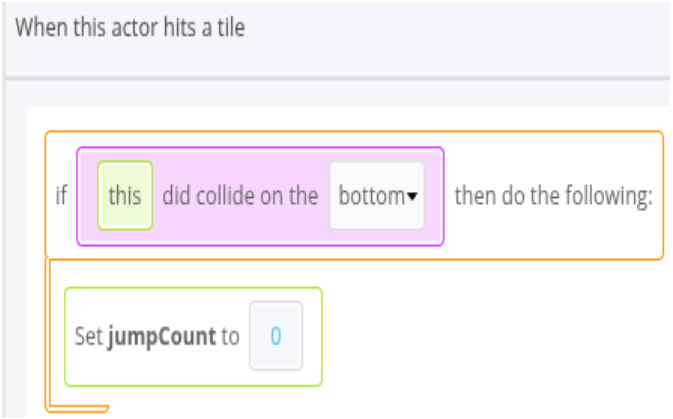
Hit save and try jumping around! Wait. We can only jump once in the entire game?

3. Define Reset function inside the Player actor

Click on menu: Ledger Throw

Click on Player, click on code tab, add a new function `Tile Reset`

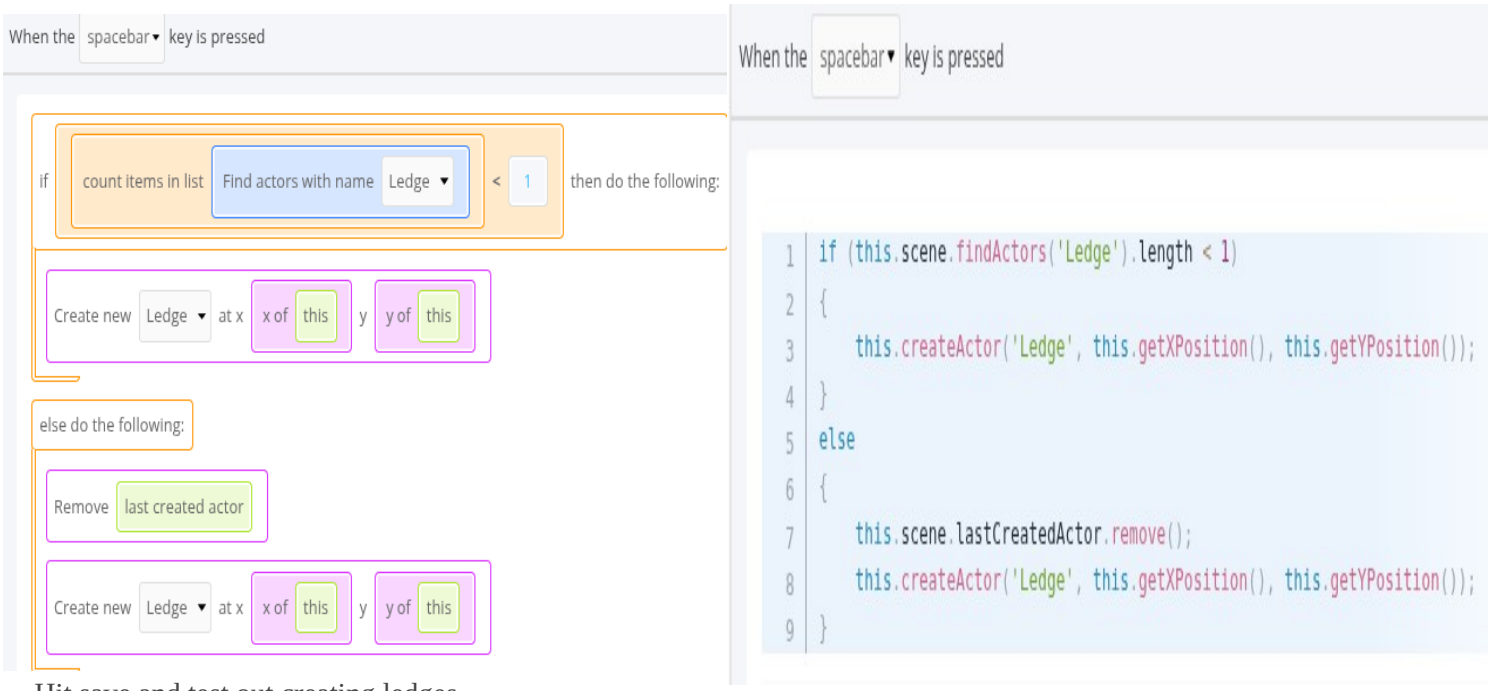




Test: Make sure you can jump up and down

4. Ledge Throwing

Add a new function Throw of Type “When a key is pressed”



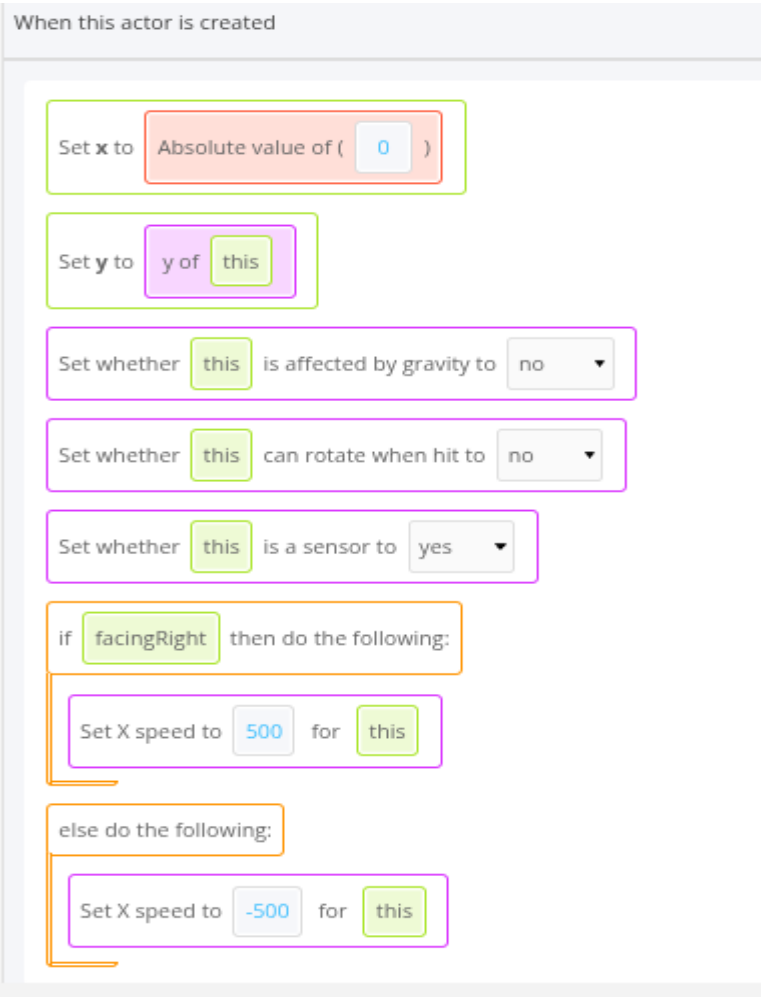
Hit save and test out creating ledges

5. The Right (and Left) Way

Now that we can create ledges, we need to make sure they fly in the right direction. To do that, we just need to use the facingRight variable. If we're facing right, we'll make it move with a speed of 500. If not, then it'll move left with a speed of 500.

We can do all that in our Ledge's **Setup** function (inside the Ledge actor)

1. Drag if from LOGIC into the code box
   - Input box: Drag Get 'facingRight' from VARIABLES into the condition box
2. Drag Set X speed from ACTOR inside the if block
   - Input box 1: Type 500 in the speed box
   - Input box 2: Drag this actor from VARIABLES inside the this block
3. Drag else from LOGIC into the code box
4. Drag Set X speed from ACTOR inside the else block
   - Input box 1: Type -500 in the speed box
   - Input box 2: Drag this actor from VARIABLES inside the this block



Save and test it out

Ledge Too Fast or Slow?

If the ledge is too fast or too slow, then you can change the number!

Instead of 500, do 1000 in the first block and -1000 in the second to make it twice as fast!

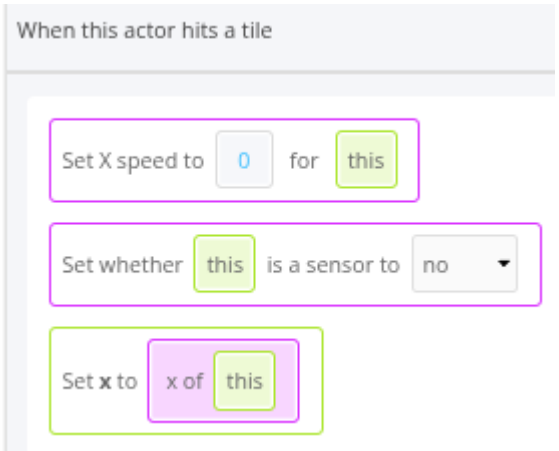
Or, instead of 500, you could do 250 in the first block and -250 in the second to make it slow it down by half!

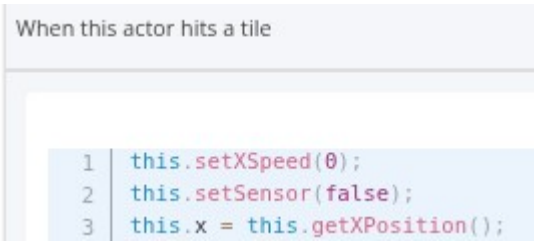
It's best not to make it too slow!

Step 6 Create a Stop function inside the Ledge actor

When our ledge hits a tile, we want it to stop so we can jump on it. We can do that by changing the x speed of the ledge.

Since our ledge is a sensor, it'd normally pass through tiles as if they weren't there. So we'll ALSO need to change it from being a sensor to no longer being one.





Save and test it out

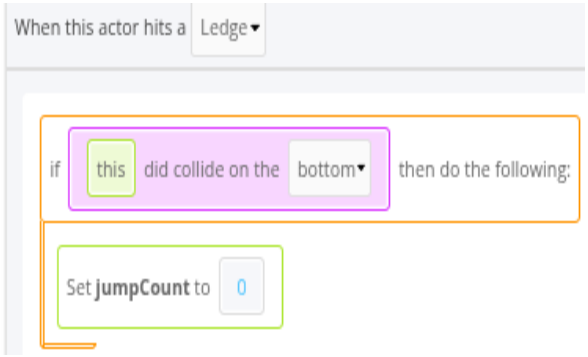
Step 7 Create a Ledge Reset function when this actor hits another actor

Jumping on the Ledge

Since we only reset our jump when we hit a tile, we can't jump on the platform. Oh no! Let's make it so that our jumps also reset whenever our actor collides with a ledge with their feet.

Drag in the following code:

1. Input box: Select Ledge ▾ from the options in the dropdown
2. Drag if from LOGIC into the code box
   - Input box: Drag Did collide on side from ACTOR into the condition box
     - Input box 1: Drag this actor from VARIABLES inside the this block
     - Input box 2: Select bottom ▾ from the options in the dropdown
3. Drag Set 'jumpCount' from VARIABLES inside the if block
   - Input box 1: Type 0 in the box



Why check if we're colliding on the bottom?

This means that the feet of our Player are touching the ledge.

If we **didn't**, then as long as any part of the Player hits a ledge, we can jump again.

That means all we'd need to do is touch it with our head and our jumps would reset! That'd make the game TOO easy!So, when we check if we collided on the bottom, we're checking that we're standing on the platform and not that we're hitting it some other way where jumping wouldn't make sense.

Step 8. Another Escape

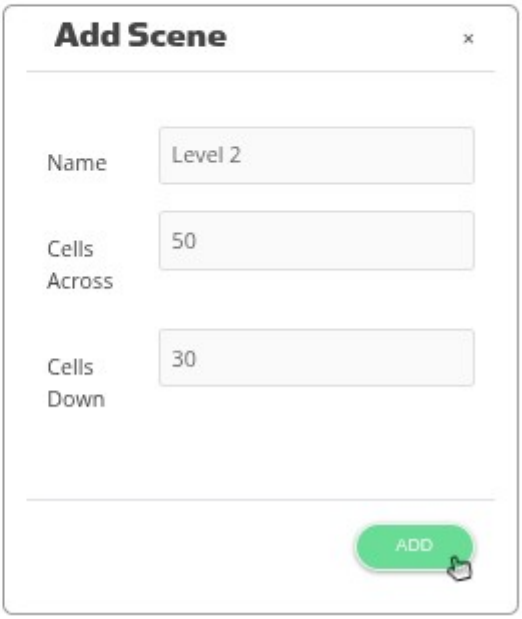
Test it out once you play. Can't see everything? Click on the three dots in the top right of the level and click **Resize**. Changing the scaling mode to **Show entire scene** makes it so everything will show!

Make sure it's more challenging than the first level but not TOO challenging!

After you've made one level, keep going! How many levels can you create?

Click on the **Add Scene** button

Create a scene with the following details:

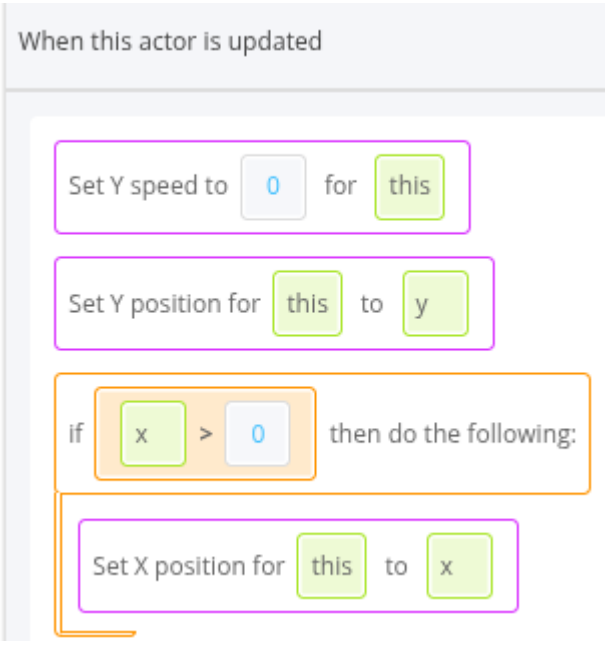


Step 9 Bug fix. Go to Stay Still function under Actor Ledge

Check **if** x is greater than 0

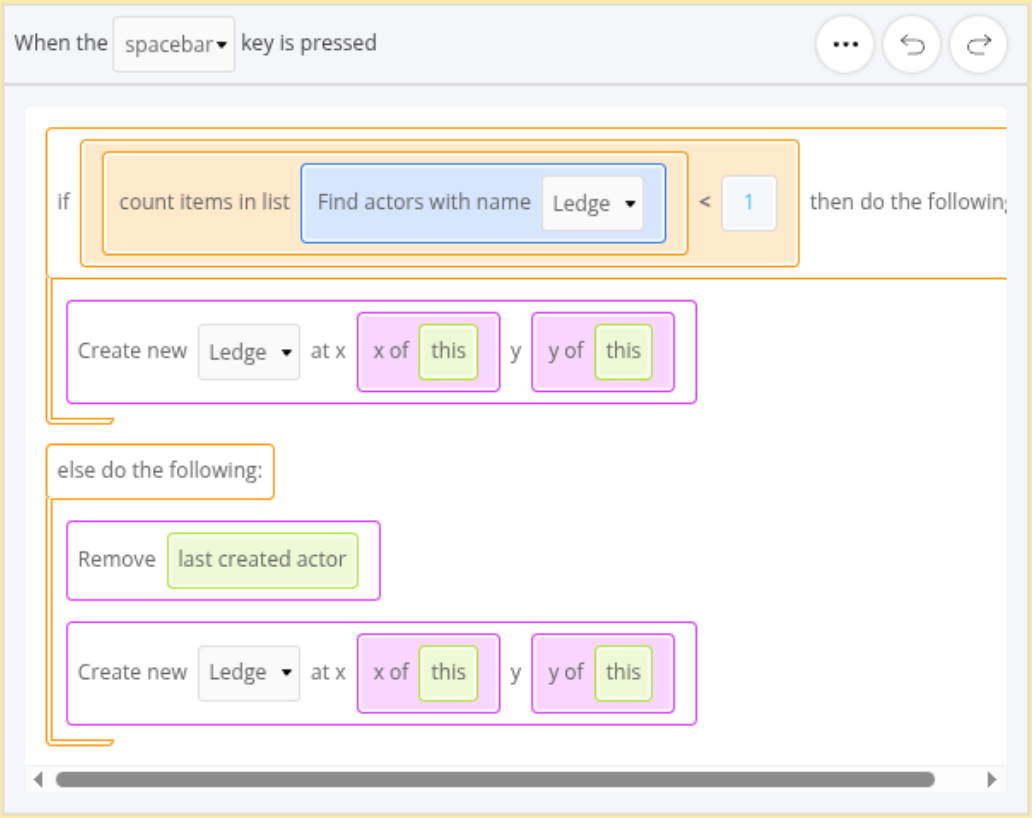
Inside the if block, set the x position of the ledge to Get 'x'

Students may have noticed you can make the ledge move left / right. We have a variable, **x**, that keeps track of the x position the ledge when it stops. We can use that to ensure it keeps that position.



Step 9 Wrap up

- **Question**: Looking at our Throw function, why do we have an else? Why can't we get rid of it?



Answer: We need to make sure that we **remove** the actor when we want to create a new one. We can't always remove the last created actor as that will either be the player or the door when you first start the level! So the if condition will only apply for the first time we throw a ledge.

Question: In our tile and ledge jump resets, why did we have to check if we collided on the bottom?

Answer: Since we can collide with a tile and actor in multiple directions, we need to make sure we're not able to cheat the system! By checking if we're colliding on the bottom, we know the bottom part of the actor is colliding with the tile or ledge. If we didn't have the check, we could just hug a wall and jump straight up or only need to touch the ledge with our head to reset our jump count.

Does anybody have any questions?